



Fermentation and Alcohol

Yeasts are alcohol fermenters: They break down sugars into carbon dioxide and ethanol. As a result, they produce both carbonation (as seen in beer and sparkling wines) and alcohol. Eventually, a combination of temperature change (perhaps introduced deliberately during a brewing process), too high an alcohol concentration, and/or a lack of available sugars will kill the yeasts, ending the fermentation process. Of course, some alcoholic beverages contain living yeasts that continue the fermentation process up to the point when you drink them!

There are two key benefits to alcohol fermentation. The first is that the presence of alcohol inhibits the growth of other microorganisms, resulting in a safe source of water and the stable preservation of fruit juice and grains that can otherwise go bad from the growth of undesirable molds and bacteria. The second benefit is the alcohol itself, a narcotic with considerable social importance across time and space, as we'll discuss in more detail in a future module.

Learn about the most common alcohol fermented beverages—beers, wines, and vinegars—here, as well as how the process of distillation fits in.

Beers

Beers are generally defined as fermented alcoholic beverages produced from the fermentation of grain. The most “typical” beers today follow (strictly or loosely) the *Reinheitsgebot*, or German “purity order” (dating back to the 15th century!), which defines the ingredients of beer as only water, yeast, hops, and malted barley. Malting is the process of sprouting a grain—which activates enzymes in the seed that begin to convert starch to sugar to feed the seedling plant—and then drying and heating the grain to kill the seedling just as it begins to sprout. The resulting grain has a much higher percentage of sugar than available to feed yeast during the brewing process. Of course, nonmalted grain can also be brewed, but the yeasts have less available food initially, and other yeast strains or microorganisms may be needed to ensure that the process works correctly. Most beers traditionally have a moderate alcohol concentration—from 4–6%—although modern yeast strains have been developed that can survive alcohol concentrations as high as 20%, allowing for very high-alcohol beers, on par with wines (~14% alcohol). Such beverages may be termed “barley wines” but are just high-alcohol beers.

While we think of beers as using barley as the primary grain, many other grains can be used, as well. One common substitute is wheat, such as is found in German *Weizenbiere* (wheat beers), but others abound. Sake is a rice beer, chicha is a maize beer, and early Chinese emperors and diviners

used millet beers to communicate with the spirit world. We'll learn more about these cultural contexts for alcohol in a future module.

Wines

Wines are generally defined as alcoholic beverages fermented from fruit. We of course refer to grape wine in the most generic use of the term today, but other fruits have commonly been used in different times and places in human history. Note that, by this definition, alcoholic ciders are also wines: They are fermented from fruit juice (apple cider, for example). However, in common English today, "cider" refers to the juice of either apples or similar fruit (e.g., pears), which may or may not have been fermented into an alcoholic beverage.

The fermentation process for wines differs from that for beers in two ways: Abundant sugar is already available from the ripe fruit, so malting or an equivalent preparation stage is not necessary, and wild yeasts are often used. Grape-wine production traditionally involves mashing ripe grapes and allowing the resultant juice to sit with the grape skins for a short period (for white wines) or a longer period (for red wines); in both cases, the yeasts that naturally grow on the grape skins as they mature are the yeasts that begin the fermentation of grape juice into wine. Today, in industrial wine production, specific grape-wine yeast strains may be introduced, as well, to further control the flavor of the wine and speed fermentation, but this is not traditional for most European wine producers.

Wines, whether of grape or other fruits, are often much more alcoholic than beers, due to the abundant sugars available. Wines sold today generally have an alcohol content of 13-15%, although some sweeter dessert wines (made from very high sugar fruit, often with an additional input of sugar) may have significantly higher alcohol concentrations, up to 20%.

Vinegar

Vinegar is a byproduct of fermentation. Vinegar is a fermented food, but indirectly: Acetic acid, the primary component of vinegar, is formed by the oxidation of alcohol. Without alcohol and oxygen, vinegar does not form. This is why poorly sealed wine or hard apple cider can "go bad," producing an acidic smell due to the conversion of alcohol into acetic acid when it is in the presence of oxygen. A properly corked bottle of wine excludes oxygen, allowing the wine to age with minimal to no oxidation.

Vinegar can be produced from any alcoholic liquid, from beers to ciders and wines. Often, the alcohol is produced solely for the purpose of vinegar production; for example, white (or distilled) vinegar is created by producing a neutral distilled spirit (similar to a vodka, but much stronger) of nearly pure alcohol, which is then oxidized to acetic acid and diluted with water to around 5% acidity (and a pH around 3) for sale. Note that none of the original alcohol remains present in most commercially sold vinegars, though some traditional vinegar-production methods may result in trace (less than 1%) alcohol content remaining.